Linear Discriminant Analysis

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Introduction

Linear discriminant analysis (LDA) is the canonical Gaussian classifier for two or more classes. Under the assumption that within-class distributions are multivariate Normal with a common covariance matrix, the optimal Bayes classifier is linear in the predictors, and the corresponding posterior probabilities are simple closed-form functions of class means and the pooled covariance. Fisher's complementary derivation gives a projection onto $K-1$ axes that maximally separates the K class centroids; this is the most useful low-dimensional visualisation for class structure.

Prerequisites

A working understanding of multivariate Normal distributions, classification metrics, and the Bayes-optimal classifier under known generative assumptions.

Theory

Assume class k has prior π_k, mean μ_k, and common covariance Σ. The Bayes-optimal log-discriminant is

$$\delta_k(x) = x^\top \Sigma^{-1}\mu_k - \tfrac{1}{2}\mu_k^\top \Sigma^{-1}\mu_k + \log \pi_k.$$

Assigning x to the class with the largest δ_k gives the LDA classifier. The decision boundary between any two classes is linear in x, which is what makes the model interpretable: classification depends on a small number of linear combinations of the predictors.

Fisher's LDA, conceptually identical, finds the directions in which the between-class variance is large relative to the within-class variance; projecting onto the top $K-1$ directions gives the canonical low-dimensional visualisation of class structure.

Assumptions

Multivariate Normal within each class, equal class covariance matrices (otherwise QDA is preferred), and independent observations. Outliers can pull class means and inflate the pooled covariance, distorting the linear boundary.

R Implementation

```r
library(MASS)

set.seed(2026)
train_idx <- sample(nrow(iris), 100)
train <- iris[train_idx, ]; test  <- iris[-train_idx, ]

fit_lda <- lda(Species ~ ., data = train)
pred <- predict(fit_lda, test)
table(pred$class, test$Species)
```

```r
# Project onto LDA axes
plot(fit_lda, col = as.integer(train$Species))
```

Output & Results

`lda()` returns class priors, class means, scaling matrix (coefficients of linear discriminants), and the proportion of trace each axis explains. `predict()` provides class labels and posterior probabilities. The classifier projects test observations onto the LDA axes and assigns each to the nearest class centroid in the projected space.

Interpretation

A reporting sentence: "Linear discriminant analysis on the iris training set classified the held-out observations with 98 % accuracy (49/50 correct); the first discriminant axis accounted for 99 % of between-group variance, indicating that a single linear combination of the four measurements largely separates the three species." Compare LDA to QDA when class covariances differ; otherwise LDA is the more efficient choice.

Practical Tips

- For class-specific covariance differences, switch to quadratic discriminant analysis (QDA); Box's M test (`biotools::boxM`) provides a formal check, but it is sensitive to non-Normality.
- LDA is sensitive to outliers; screen for them or use robust variants such as `MASS::lda(... , method = "mve")` or `rrcov::Linda()`.
- Prior class probabilities affect the decision boundary; specify them via `prior = c(...)` when training-set proportions differ from population proportions.
- For high-dimensional data with p approaching or exceeding n, regularise via `klaR::rda()` (regularised discriminant analysis) or sparse LDA (`MASS::lda` with PCA pre-step, or `sparseLDA`).
- Multiclass alternatives include multinomial logistic regression (`nnet::multinom`) and random forests; LDA remains competitive in low-dimensional problems with Gaussian-like classes.
- Visualise the LDA projection with the first two discriminants; a 2D plot with class colours quickly shows whether classes are linearly separable.

R Packages Used

`MASS` for `lda()` and `qda()`; `klaR` for `rda()` (regularised) and `partimat()` for partition diagnostics; `caret` for cross-validated comparison across discriminant variants; `rrcov` for robust LDA; `sparseLDA` for high-dimensional sparse extensions.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- PCA, factor analysis, CCA, LDA
- Clustering: k-means, hierarchical, model-based
- UMAP and t-SNE